

Science — Cbse Annual Exam — 80 Marks

Subject: Science | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	Local winds like the mountain breeze influence weather conditions and daily life in mountainous regions. Which of the following is a benefit of these local winds? (A) They increase atmospheric CO₂ levels. (B) They accelerate the melting of glaciers. (C) They cause ocean currents to reverse direction. (D) They regulate temperature and moisture conditions, supporting soil and crop health. <i>[Ch 13: Earth as a System: Energy, Matter, and Life]</i>	1
2	When two forces are opposite in direction but unequal in magnitude, what is the magnitude of the net force? (A) The sum of the magnitudes of the two forces (B) The difference between the magnitudes of the two forces (C) The average of the magnitudes of the two forces (D) Zero <i>[Ch 6: How Forces Affect Motion]</i>	1
3	Who proposed that any living plant cell can develop into a complete plant if provided with suitable nutrients and favourable conditions? (A) Arun Kumar Sharma (B) Theodor Schwann (C) Rudolf Virchow (D) Gottlieb Haberlandt <i>[Ch 2: Cell: The Building Block of Life]</i>	1



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4	In a game of carrom, a moving striker hits a stationary white coin. What happens to the energy of the white coin? (A) It decreases because the striker does negative work on it (B) It increases because the striker does positive work on it (C) It remains constant because no work is done (D) It becomes zero upon collision <i>[Ch 7: Work, Energy, and Simple Machines]</i>	1
5	What are the special organelles found in plant cells that include chloroplasts, leucoplasts, and chromoplasts? (A) Plastids (B) Lysosomes (C) Ribosomes (D) Endoplasmic reticulum <i>[Ch 2: Cell: The Building Block of Life]</i>	1
6	Assertion (A): The formation of different types of tissues leads to division of labour in multicellular organisms. Reason (R): Different groups of cells perform different functions, which increases the efficiency of the body and enables it to carry out complex life processes. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 3: Tissues in Action]</i>	1
7	Assertion (A): The divisions of science into branches like physics, chemistry, biology and earth science are independent of each other. Reason (R): Most real-world problems today, such as understanding climate change, developing medicines, or designing sustainable technologies, require ideas from several disciplines together. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1
8	Which joint allows forward, backwards, sideways, and circular movements? (A) Hinge joint (B) Pivot joint (C) Fixed joint (D) Ball and socket joint <i>[Ch 3: Tissues in Action]</i>	1



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9	The claim that food becomes harmful during an eclipse is best disproved by asking which type of question? (A) Whether people have believed this claim for centuries (B) Whether the eclipse is total or partial (C) Whether any physical, chemical, or biological mechanism supports the claim (D) Whether ancient texts mention this practice <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1
10	After a zygote is formed, it undergoes a series of mitotic divisions while travelling to the uterus and then: (A) Is released through the vagina (B) Returns to the ovary (C) Degenerates within a day (D) Implants into the inner lining of the uterus <i>[Ch 11: Reproduction: How Life Continues]</i>	1
11	Which of the following is an example of a colloid? (A) Milk (B) Salt water (C) Sand in water (D) Sugar in water <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1
12	What is the primary role of the dermal tissue system in plants? (A) Transporting water and minerals throughout the plant (B) Forming the main body of the plant between dermal and conducting tissues (C) Forming the outer covering of the plant, protecting inner parts and reducing water loss (D) Increasing the length and girth of the plant <i>[Ch 3: Tissues in Action]</i>	1
13	Given that the atomic mass of K = 39 u and Cl = (A) 35.5 u (B) 39 u (C) 3.5 u (D) 74.5 u <i>[Ch 9: Atomic Foundations of Matter]</i>	1



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14	Brass is an alloy composed of approximately: (A) 80% zinc and 20% copper (B) 50% copper and 50% zinc (C) 80% copper and 20% zinc (D) 80% copper and 20% tin <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1
15	Which of the following statements about loudness and intensity is correct? (A) Loudness and intensity are exactly the same quantity (B) Intensity is a measurable quantity, whereas loudness depends on the listener's hearing ability (C) Intensity is subjective, while loudness is measurable (D) Loudness is independent of amplitude <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	1
16	In the airplane fuel miscalculation incident, what was the cause of the error? (A) The fuel tank had a leakage mid-flight (B) The pilot misread the fuel gauge instruments (C) The density of fuel was used in pounds per litre instead of kilograms per litre (D) The aircraft was overloaded with passengers and cargo <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1
17	Assertion (A): The sound of submerged metal spoons tapped under water reaches the listener after travelling through both water and air. Reason (R): Sound can only travel through a single medium at a time. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	1
18	How does the elongated, cylindrical body structure of roundworms benefit them? (A) It allows them to fly efficiently (B) It helps them produce their own food (C) It reduces water loss in dry environments (D) It allows efficient movement through soil, water, or host tissues <i>[Ch 12: Patterns in Life: Diversity and Classification]</i>	1



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19	Assertion (A): The ovary of a flower develops into a fruit after fertilisation. Reason (R): The ovules inside the ovary develop into seeds. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 11: Reproduction: How Life Continues]</i>	1
20	Assertion (A): Most cellular activities in prokaryotic cells take place directly in the cytoplasm. Reason (R): Prokaryotic cells lack membrane-bound organelles. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Cell: The Building Block of Life]</i>	1
Section B Attempt 5 out of 7.		
21	What is the purpose of using models in science, and what do they involve? <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	2
	OR In a simple pendulum, why does the bob eventually stop oscillating instead of reaching the same height indefinitely? <i>[Ch 7: Work, Energy, and Simple Machines]</i>	
22	Fill in the blanks. Among magnesium and chlorine, magnesium atom can give two electrons to become Mg^{2+}. However, chlorine can take only one electron to become _____. Now, _____ ion of magnesium and _____ ions of chlorine combine to give magnesium chloride. <i>[Ch 9: Atomic Foundations of Matter]</i>	2
	OR How does the magnitude of the force of friction affect the distance travelled by an object before coming to rest? <i>[Ch 6: How Forces Affect Motion]</i>	
23	Why are the epithelial tissues lining an animal's internal organs usually only one or a few cells thick? <i>[Ch 3: Tissues in Action]</i>	2
24	Define isotonic, hypotonic, and hypertonic solutions based on solute concentration. <i>[Ch 2: Cell: The Building Block of Life]</i>	2



Q.NO.	QUESTIONS	MARKS
25	Differentiate between sublimation and deposition based on the state changes involved. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	2
Section C Attempt 5 out of 7.		
26	A perfume mixture is prepared by mixing 15 mL of fragrance oil with enough water to make a total volume of 75 mL of solution. Calculate the volume by volume percentage (% v/v) of the fragrance oil in the mixture. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	3
	OR Explain how the structure of the inner membrane of a mitochondrion is well-adapted to facilitate energy production. <i>[Ch 2: Cell: The Building Block of Life]</i>	
27	State the principle of the lever using the relationship between effort, load, and their respective arms. How does increasing the effort arm affect the effort required? <i>[Ch 7: Work, Energy, and Simple Machines]</i>	3
	OR Describe the arrangement of ions in a sodium chloride crystal and explain what a crystal lattice is. <i>[Ch 9: Atomic Foundations of Matter]</i>	
28	An aircraft ran out of fuel mid-flight because the ground crew used the density of fuel in pounds per litre instead of kilograms per litre. What does this incident highlight about the importance of using standard units (SI)? <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	3
29	Differentiate between the apical, lateral, and intercalary meristems based on their location and function in plants. <i>[Ch 3: Tissues in Action]</i>	3
30	According to Newton's second law of motion, how does the acceleration of an object relate to the net force acting on it and its mass? <i>[Ch 6: How Forces Affect Motion]</i>	3
Section D Attempt 4 out of 6.		
31	Differentiate between laws, theories, and principles in science. Explain each term with an example and discuss why a theory in science is not the same as a guess. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	5



Q.NO.	QUESTIONS	MARKS
	OR Describe the process of meiosis and explain its significance in living organisms. Where does meiosis occur in animals and plants, and how is the original chromosome number restored after meiosis? <i>[Ch 2: Cell: The Building Block of Life]</i>	
32	How would you separate the mixtures given in Table 5.3? Mention the reason for choosing your method. If a mixture cannot be separated, explain why. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	5
	OR A cricket fielder gradually pulls their hands backwards while catching a fast-moving ball, and airbags are provided in vehicles. Explain how Newton's second law of motion is applied in both these situations to reduce the force experienced. Also, explain why a coconut brought down at high velocity against a hard surface experiences a very large force. <i>[Ch 6: How Forces Affect Motion]</i>	
33	Describe the structural characteristics and types of meristematic tissues in plants. Explain how these tissues give rise to permanent tissues. <i>[Ch 3: Tissues in Action]</i>	5
34	Describe the energy transformations that take place in a traditional water mill (gharat) used in the Himalayan region to grind grain. How is this same principle applied in modern times to generate electricity? <i>[Ch 7: Work, Energy, and Simple Machines]</i>	5
Section E Attempt 3 out of 4.		
35	Case Study: Observing Cells Under a Microscope Rohan observed onion peel cells under a microscope. He counted 25 cells along the diameter of the visible field, which was 5000 μm. His microscope had an eyepiece and an objective lens, each with a magnifying power of 10X. He noticed the cell membrane formed a distinct boundary around each cell. He recalled that the cell membrane is composed of proteins and lipids, and that it serves as the boundary where substances move between the cell and its external environment. He also learned that microscopes are improved by enhancing three main features: resolution, contrast, and magnification. (i) Calculate the estimated size of one onion peel cell based on the data provided. (ii) What is the total magnification of Rohan's microscope? How many times larger will the cell appear under this magnification? (iii) The cell membrane is composed of proteins and lipids. Describe the role of the cell membrane as a boundary between the cell and its external environment. (iv) Among the three main features of a microscope—resolution, contrast, and magnification—which one is described as the 'measure of clarity,' and why is improving this feature important for studying cell structures?	5



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	<i>[Ch 2: Cell: The Building Block of Life]</i>	
	OR Case Study: Building Scientific Models A student wants to model how long it takes for a ball thrown upward to return to the ground. She considers several factors: the mass of the ball, its initial speed and direction, air resistance, the colour of the ball, and the spin of the ball. She recalls that physicist Meghnad Saha simplified the study of stars by ignoring complex processes and focusing only on temperature, pressure, and how atoms formed ions, which helped explain the connection between star colour and temperature. She decides to apply a similar simplification approach. She also notes that science uses precise symbols like m for mass, v for velocity, and F for force, and that equations are compact statements about relationships between quantities. (i) What three factors did Meghnad Saha focus on to explain the colour of stars? (ii) Why should the student ignore the colour of the ball when modeling the time it takes to return to the ground? (iii) The student ignores air resistance and spin in her simple model. Later, she wants a more accurate prediction. What should she do, based on the approach described for building complex models? (iv) The student writes an equation relating distance, time, and velocity to describe the ball's motion. Explain why this equation is more than just a calculation tool. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	
36	Case Study: Concentration of Solutions in Everyday Products A student examines three products. A packaged food item contains 8 g of salt in 200 g of the product. A glucose solution bottle from a pharmacy is labelled 5% m/v, meaning 5 g of glucose is present in 100 mL of the solution. A milk powder packet lists its protein content as 15% m/m. The student notes that mass by mass percentage is commonly used for heterogeneous mixtures like milk powder, while mass by volume percentage is used in medicines and laboratories where measuring the volume of a liquid is easier than weighing it. (i) What does 5% m/v on the glucose solution bottle indicate? (ii) Why is mass by volume percentage used for medicines like glucose solution instead of mass by mass percentage? (iii) Calculate the mass by mass percentage of salt in the packaged food item if it contains 8 g of salt in 200 g of the product. (iv) Milk powder is labelled using mass by mass percentage for its composition. What type of mixture is milk powder, and why is this method of expressing concentration appropriate for it? <i>[Ch 5: Exploring Mixtures and their Separation]</i>	5



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37	Case Study: Totipotency and Tissue Culture In 1958, F.C. Steward demonstrated that phloem cells from carrot could regenerate whole plants when grown in a nutrient medium containing simple sugars and hormones. The phloem cells first dedifferentiated into an undifferentiated mass of unspecialised cells, then redifferentiated into roots, shoot, and a complete plant. His experiments showed that cells in liquid medium with nutrients and air showed 20% increase in fresh weight, while cells in solid medium under light, or in liquid medium without air (condition X), showed reduced growth. Separately, scientists found that <i>Agrobacterium tumefaciens</i> causes crown gall disease, creating tumour-like swellings from rapid, uncontrolled cell division, and this bacterium is now used to introduce useful genes into plants. (i) Define totipotency. (ii) In Steward's experiment, cells in liquid medium with air showed 20% increase in fresh weight, while cells in condition X (liquid medium with nutrients) showed reduced growth. What does condition X lack, and why did growth reduce? (iii) A scientist cultures phloem cells from a carrot plant in a nutrient medium with simple sugars, hormones, and air. Describe the complete sequence of cellular changes that would enable these cells to form a whole plant. (iv) <i>Agrobacterium tumefaciens</i> causes crown gall disease through rapid and uncontrolled cell division. Analyze how studying this disease mechanism led to a beneficial application in plant science. <i>[Ch 3: Tissues in Action]</i>	5

